Heteroencoding

Baseline Experiment

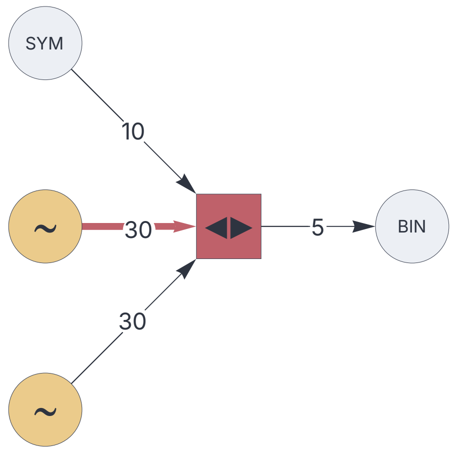
Testing heteroencoding with a synthetic dataset, consisting of random integers from 0 to 9 encoded as random SHRs with population 10. This circuit adds 30 random elements and randomly removes a few elements.
The heteroencoder component successfully clusters this data into 10 bins.

Configuration

```
<< "Circuits.m"
```

```
dataflow =
    {
    random[][11][],
    11 → {1000, 30},
    random[][12][],
    12 → {1000, 30},
    input[codec] [21] [],
    21 → {1000, 10},
    heteroencoder[] [101] [{11"%",12 "-%", 21"%"}],
    output[binning] [] [101"%"],
    101 → {500, 5}
    };
```

```
c = Circuit[ dataflow]; c[Graph, ImageSize→ 240]
```



Synthetic data generation

```
dataset = RandomInteger[ {0, 9}, 10000];
```

Train

```
bins = c /@ dataset;
```

Analyze

```
SortBy[Tally[dataset], Last]
```

```
{{7, 936}, {3, 955}, {9, 957}, {6, 962}, {5, 992},
 {0, 1005}, {2, 1027}, {8, 1032}, {4, 1051}, {1, 1083}}
```

```
SortBy[Tally[bins], Last]
```

```
{{7, 936}, {10, 955}, {5, 957}, {9, 962}, {4, 992},
 {8, 1005}, {1, 1027}, {3, 1032}, {2, 1051}, {6, 1083}}
```

The algorithm has identified 10 clusters and correctly classified the data.